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BATCH:26

Assignment Number:20.5		
Q.No.	Question	Expected Time to complete
1	Lab 20 – Security Testing: Identifying Vulnerabilities in AI-Generated Code Lab Objectives: • Understand how to test AI-generated code for common security vulnerabilities. • Learn to apply secure coding principles while analyzing AI outputs. • Practice detecting risks such as SQL injection, XSS, hardcoded credentials, and weak encryption. • Enhance code reliability and safety by using AI for secure refactoring.	Week10 - Friday
	Task 1 – Password Storage Vulnerability Check Task: Analyze an AI-generated signup script for insecure password storage. Instructions: • Prompt AI to generate a Python user registration program. • Check whether passwords are stored in plain text. • Improve the script using hashing (bcrypt or hashlib). Expected Output: • A secure registration script with hashed password storage.	

```

AAC A205py X
C:\Users\kruth> python A205py.py
1 import json
2 import os
3 import hashlib
4 from pathlib import Path
5
6 DB_FILE = Path("users.json")
7 ITERATIONS = 200_000
8
9 def load_users():
10     if not DB_FILE.exists():
11         return {}
12     with DB_FILE.open("r", encoding="utf-8") as f:
13         return json.load(f)
14
15 def save_users(users):
16     with DB_FILE.open("w", encoding="utf-8") as f:
17         json.dump(users, f, indent=2)
18
19 def hash_password(password, salt=None):
20     if salt is None:
21         salt = os.urandom(16)
22     dk = hashlib.pbkdf2_hmac("sha256", password.encode("utf-8"), salt, ITERATIONS)
23     return salt.hex(), dk.hex()
24
25 def register_user(username, password):
26     users = load_users()
27     if username in users:
28         return False, "Username already exists."
29     salt_hex, hash_hex = hash_password(password)
30     users[username] = {
31         "salt": salt_hex,
32         "password_hash": hash_hex,
33         "iterations": ITERATIONS,
34     }
35     save_users(users)
36     return True, "User registered successfully."
37
38 if __name__ == "__main__":
39     username = input("Enter username: ")
40     password = input("Enter password: ")
41     ok, message = register_user(username, password)
42     print(message)
43
44 PS C:\Users\kruth> python A205py.py
Enter username: Dhruvhan Kiran
Enter password: Kruthan@100
User registered successfully.
PS C:\Users\kruth>

```

```

19 def hash_password(password, salt=None):
20     if salt is None:
21         salt = os.urandom(16)
22     dk = hashlib.pbkdf2_hmac("sha256", password.encode("utf-8"), salt, ITERATIONS)
23     return salt.hex(), dk.hex()
24
25 def register_user(username, password):
26     users = load_users()
27     if username in users:
28         return False, "Username already exists."
29     salt_hex, hash_hex = hash_password(password)
30     users[username] = {
31         "salt": salt_hex,
32         "password_hash": hash_hex,
33         "iterations": ITERATIONS,
34         "algorithm": "pbkdf2_hmac_sha256"
35     }
36     save_users(users)
37     return True, "User registered successfully."
38
39 def verify_user(username, password):
40     users = load_users()
41     user = users.get(username)
42     if not user:
43         return False
44     salt = bytes.fromhex(user["salt"])
45     iterations = int(user.get("iterations", ITERATIONS))
46     dk = hashlib.pbkdf2_hmac("sha256", password.encode("utf-8"), salt, iterations)
47     return dk.hex() == user["password_hash"]
48
49 if __name__ == "__main__":
50     username = input("Enter username: ").strip()
51     password = input("Enter password: ").strip()
52     ok, message = register_user(username, password)
53     print(message)
54
55 PS C:\Users\kruth> python A205py.py
Enter username: Dhruvhan Kiran
Enter password: Kruthan@100
User registered successfully.
PS C:\Users\kruth>

```

Task 2 – Hardcoded Secrets Detection

Task:

Evaluate AI-generated code for hardcoded API keys or credentials.

Instructions:

- Ask AI to generate a script that connects to an external API.
- Identify if API keys/passwords are written directly in code.
- Refactor using environment variables or config files.

Expected Output:

- A secure script with secrets managed properly.

```
1 import os
2 import json
3 from pathlib import Path
4 import requests
5
6 BASE_URL = os.getenv("WEATHER_API_BASE_URL", "https://api.openweathermap.org/data/2.5/weather")
7 CONFIG_PATH = Path(__file__).with_name("weather_config.json")
8
9 def load_api_key() -> str | None:
10     api_key = os.getenv("WEATHER_API_KEY")
11     if not api_key:
12         return None
13     return api_key
14
15 if CONFIG_PATH.exists():
16     with CONFIG_PATH.open("r", encoding="utf-8") as f:
17         data = json.load(f)
18         key = data.get("WEATHER_API_KEY")
19         if key:
20             return key
21         return None
22
23 def get_weather(city: str) -> dict:
24     api_key = load_api_key()
25     if not api_key:
26         api_key = input("Enter WEATHER_API_KEY: ").strip()
27     if not api_key:
28         raise RuntimeError("Missing WEATHER_API_KEY in environment, weather_config.json, and runtime input")
29     params = {"q": city, "appid": api_key, "units": "metric"}
30     response = requests.get(BASE_URL, params=params, timeout=15)
31     if response.status_code == 401:
32         raise RuntimeError("Invalid WEATHER_API_KEY. Please verify or generate a new key.")
33     response.raise_for_status()
34     return response.json()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS

PS C:\Users\kruth> & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '-.' 'c:\Users\kruth\AAC A 20.5.py'

requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c6269619ae6348e093b0aa4209a

PS C:\Users\kruth> cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '-.' 'c:\Users\kruth\AAC A 20.5.py'

City: Parkal

Enter WEATHER_API_KEY: 84f61c6269619ae6348e093b0aa4209a

Invalid WEATHER_API_KEY. Please verify or generate a new key.

```
21 def get_weather(city: str) -> dict:
22     api_key = load_api_key()
23     if not api_key:
24         api_key = input("Enter WEATHER_API_KEY: ").strip()
25     if not api_key:
26         raise RuntimeError("Missing WEATHER_API_KEY in environment, weather_config.json, and runtime input")
27     params = {"q": city, "appid": api_key, "units": "metric"}
28     response = requests.get(BASE_URL, params=params, timeout=15)
29     if response.status_code == 401:
30         raise RuntimeError("Invalid WEATHER_API_KEY. Please verify or generate a new key.")
31     response.raise_for_status()
32     return response.json()
33
34 if __name__ == "__main__":
35     try:
36         city = input("City: ").strip()
37         data = get_weather(city)
38         print(data)
39     except requests.exceptions.RequestException as e:
40         print(f"Network/API error: {e}")
41     except RuntimeError as e:
42         print(e)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS

PS C:\Users\kruth> & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '-.' 'c:\Users\kruth\AAC A 20.5.py'

requests.exceptions.HTTPError: 401 Client Error: Unauthorized for url: https://api.openweathermap.org/data/2.5/weather?q=Parkal&appid=84f61c6269619ae6348e093b0aa4209a

PS C:\Users\kruth> cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '55964' '-.' 'c:\Users\kruth\AAC A 20.5.py'

City: Parkal

Enter WEATHER_API_KEY: 84f61c6269619ae6348e093b0aa4209a

Invalid WEATHER_API_KEY. Please verify or generate a new key.

Task 3 – File Upload Security Validation

Task:

Test an AI-generated file upload program for unsafe file handling.

Instructions:

- Prompt AI to generate a Flask file upload example.
- Check for vulnerabilities like uploading executable files.
- Add validation for file type, size, and secure filename handling.

Expected Output:

- A secure file upload implementation.

```

C:\Users\kash> cd C:\Users\kash\ > AAC A 20.5.py
1 from flask import Flask, request, jsonify
2 from werkzeug.utils import secure_filename
3 from pathlib import Path
4 import uuid
5
6 app = Flask(__name__)
7 app.config['MAX_CONTENT_LENGTH'] = 5 * 1024 * 1024
8
9 UPLOAD_DIR = Path('uploads')
10 UPLOAD_DIR.mkdir(exist_ok=True, exist_ok=True)
11
12 ALLOWED_EXTENSIONS = {'png', 'jpg', 'jpeg', 'pdf', 'txt'}
13 EXTENSION_TO_SIZE = {
14     'png': ('png'),
15     'jpg': ('jpg'),
16     'jpeg': ('jpeg'),
17     'pdf': ('pdf'),
18     'txt': ('txt'),
19 }
20
21 def detect_kind(data: bytes) -> str:
22     if data.startswith(b'\xff\xd9\xff'):
23         return 'png'
24     if data.startswith(b'\x2550a1'):
25         return 'jpg'
26     if data.startswith(b'GIF87a'):
27         return 'gif'
28     if data.startswith(b'PK'):
29         return 'zip'
30     if data.startswith(b'PK'):
31         return 'zip'
32     if b'PDF' in data:
33         return 'pdf'
34     return 'unknown'
35
36 if __name__ == '__main__':
37     app.run(debug=True)
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```
AAC A 20.5.py 2 X
C:\Users> kruth > AAC A 20.5.py > ...
21 def detect_kind(data: bytes) -> str:
34     try:
35         data.decode("utf-8")
36         return "txt"
37     except UnicodeDecodeError:
38         return "unknown"
39
40 @app.post("/upload")
41 def upload_file():
42     if "file" not in request.files:
43         return jsonify({"error": "No file part"}), 400
44
45     file = request.files["file"]
46     if file.filename is None or file.filename.strip() == "":
47         return jsonify({"error": "No selected file"}), 400
48
49     original_name = secure_filename(file.filename)
50     if original_name == "":
51         return jsonify({"error": "Invalid filename"}), 400
52
53     ext = Path(original_name).suffix.lower().lstrip(".")
54     if ext not in ALLOWED_EXTENSIONS:
55         return jsonify({"error": "File extension not allowed"}), 400
56
57     head = file.stream.read(8192)
58     if not head:
59         return jsonify({"error": "Empty file"}), 400
60
61     kind = detect_kind(head)
62     if kind in ("exe", "zip", "binary", "unknown"):
63         return jsonify({"error": "Unsafe or unsupported file content"}), 400
64
65     if kind not in EXTENSION_TO_KIND[ext]:
66         return jsonify({"error": "File content does not match extension"}), 400
67
68     file.stream.seek(0)
69     stored_name = f"{uuid.uuid4().hex}_{original_name}"
70     destination = UPLOAD_DIR / stored_name
71     file.save(destination)
72
73     return jsonify({"message": "Upload successful", "file": stored_name}), 201
74
75 @app.errorhandler(413)
76 def file_too_large(_):
77     return jsonify({"error": "File too large. Max size is 5 MB"}), 413
78
79 if __name__ == "__main__":
80     app.run(debug=False)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

PS C:\Users\kruth> c: cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-ebugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '57474' '--' 'c:\Users\kruth\AAC A 20.5.py'
ModuleNotFoundError: No module named 'flask'

PS C:\Users\kruth> c: cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-ebugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
* Serving Flask app 'AAC A 20.5.'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit

```
AAC A 20.5.py 2 X
C:\Users> kruth > AAC A 20.5.py > ...
41 def upload_file():
50     if original_name == "":
51         return jsonify({"error": "Invalid filename"}), 400
52
53     ext = Path(original_name).suffix.lower().lstrip(".")
54     if ext not in ALLOWED_EXTENSIONS:
55         return jsonify({"error": "File extension not allowed"}), 400
56
57     head = file.stream.read(8192)
58     if not head:
59         return jsonify({"error": "Empty file"}), 400
60
61     kind = detect_kind(head)
62     if kind in ("exe", "zip", "binary", "unknown"):
63         return jsonify({"error": "Unsafe or unsupported file content"}), 400
64
65     if kind not in EXTENSION_TO_KIND[ext]:
66         return jsonify({"error": "File content does not match extension"}), 400
67
68     file.stream.seek(0)
69     stored_name = f"{uuid.uuid4().hex}_{original_name}"
70     destination = UPLOAD_DIR / stored_name
71     file.save(destination)
72
73     return jsonify({"message": "Upload successful", "file": stored_name}), 201
74
75 @app.errorhandler(413)
76 def file_too_large(_):
77     return jsonify({"error": "File too large. Max size is 5 MB"}), 413
78
79 if __name__ == "__main__":
80     app.run(debug=False)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

PS C:\Users\kruth> c: cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-ebugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '57474' '--' 'c:\Users\kruth\AAC A 20.5.py'
ModuleNotFoundError: No module named 'flask'

PS C:\Users\kruth> c: cd 'c:\Users\kruth'; & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-ebugpy-2025.18.0-win32-x64\bundle\libs\debugpy\launcher' '57510' '--' 'c:\Users\kruth\AAC A 20.5.py'
* Serving Flask app 'AAC A 20.5.'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit

Task 4 – Command Injection Prevention

Task:

Analyze an AI-generated Python script that executes system commands.

Instructions:

- Ask AI to generate code using `os.system()` or `subprocess`.
- Identify command injection risks with user input.
- Fix using safe subprocess calls and input validation.

Expected Output:

- A secure command execution script.

```

AAC A 20.5.py
1 import re
2 import subprocess
3
4 ALLOWED_COMMANDS = [
5     "ping", "nslookup", ["-n", "-w"], "time", ["-c", "-W"],
6     "nslookup", ["-w", "-n"], "time", ["-c", "-W"],
7 ]
8
9 SAFE_TARGET_PATTERN = re.compile("^([a-zA-Z0-9.-]{1,253})$")
10
11 def validate_target(target: str) -> str:
12     value = target.strip()
13     if not SAFE_TARGET_PATTERN.fullmatch(value):
14         raise ValueError("Invalid target")
15     if value.startswith(".") or value.endswith(".") or "." in value:
16         raise ValueError("Invalid target")
17     return value
18
19 def validate_count(count: str) -> int:
20     value = int(count)
21     if value < 1 or value > 10:
22         raise ValueError("Count must be between 1 and 10")
23     return value
24
25 def run_safe(command: str, target: str, count: int = 4) -> str:
26     cmd = command.strip().lower()
27     if cmd not in ALLOWED_COMMANDS:
28         raise ValueError("Unsupported command")
29     safe_target = validate_target(target)
30     safe_count = validate_count(str(count))
31     if cmd == "ping":
32         args = ["ping", "-n", str(safe_count), safe_target]
33     else:
34         args = ["nslookup", safe_target]
35
36     result = subprocess.run(args, capture_output=True, text=True, timeout=15, shell=False, check=False)
37     output = result.stdout if result.stdout else result.stderr
38     return output.strip()
39
40 if __name__ == "__main__":
41     user_command = input("Command (ping/nslookup): ")
42     user_target = input("Target (domain or host): ")
43     user_count = input("Ping count (1-10, default 4): ").strip() or "4"
44     try:
45         print(run_safe(user_command, user_target, int(user_count)))
46     except Exception as e:
47         print(f"Error: {e}")

```

```

PS C:\Users\kruth> & 'c:\Users\kruth\anaconda3\envs\kiran\python.exe' 'c:\Users\kruth\.vscode\extensions\ms-python...
ebupgy\launcher' '64924' '-' 'c:\Users\kruth\AAC A 20.5.py'
Reply from 142.250.182.78: bytes=32 time=22ms TTL=116
Reply from 142.250.182.78: bytes=32 time=21ms TTL=116
Reply from 142.250.182.78: bytes=32 time=22ms TTL=116

Ping statistics for 142.250.182.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 130ms, Average = 48ms
PS C:\Users\kruth>

```

Task 5 – Secure Input Handling in Web Forms

Task:

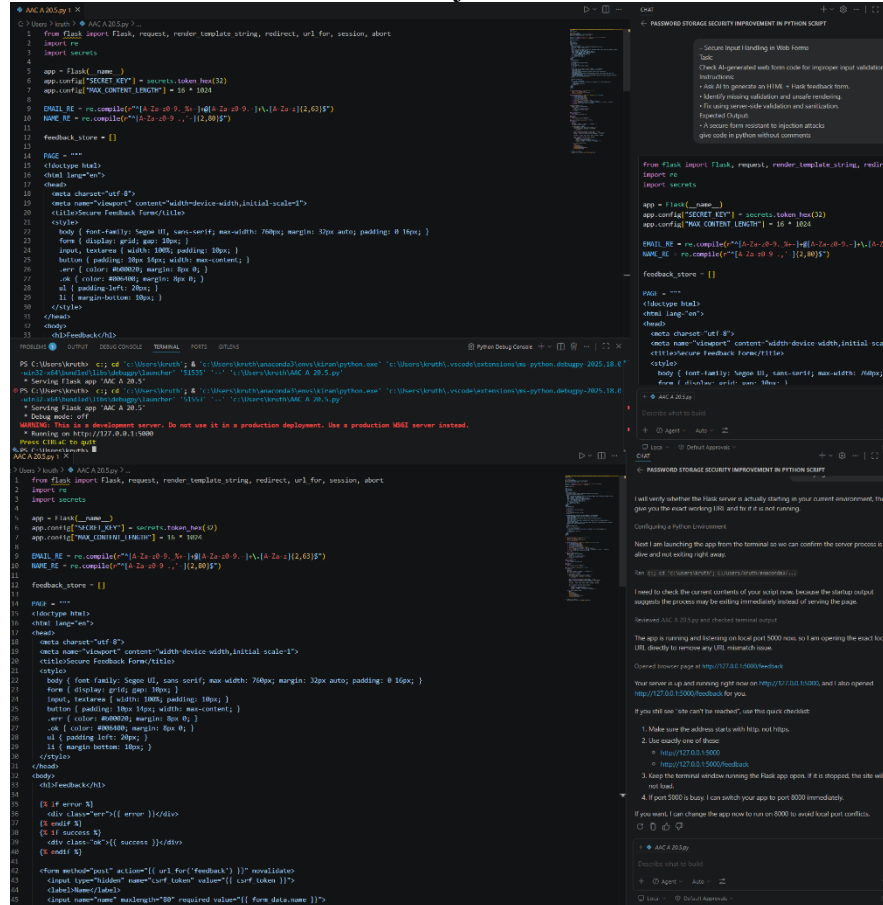
Check AI-generated web form code for improper input validation.

Instructions:

- Ask AI to generate an HTML + Flask feedback form.
- Identify missing validation and unsafe rendering.
- Fix using server-side validation and sanitization.

Expected Output:

- A secure form resistant to injection attacks.



```
1 from flask import Flask, request, render_template_string, redirect, url_for, session, secrets
2 import re
3 import secrets
4
5 app = Flask(__name__)
6 app.config["SECRET_KEY"] = secrets.token_hex(32)
7 app.config["MAX_CONTENT_LENGTH"] = 16 * 1024
8
9 EMAIL_RE = re.compile(r"^[a-zA-Z0-9.+-]+@[a-zA-Z0-9.+-]+\.[a-zA-Z]{2,6}$")
10 NAME_RE = re.compile(r"^[a-zA-Z0-9.+-]{1,20}$")
11
12 feedback_store = []
13
14 PAGE = """
15 <doctype html>
16 <html lang="en">
17 <head>
18   <meta charset="utf-8">
19   <meta name="viewport" content="width=device-width,initial-scale=1">
20   <title>Secure Feedback Form</title>
21   <style>
22     body { font-family: Segoe UI, sans-serif; max-width: 700px; margin: 10px auto; padding: 0 10px; }
23     form { display: grid; gap: 10px; }
24     input, textarea { width: 100%; padding: 10px; }
25     button { padding: 10px 15px; width: max-content; }
26     err { color: #800000; margin-top: 5px; }
27     ok { color: #006400; margin-top: 5px; }
28     ul { padding-left: 10px; }
29     li { margin-bottom: 10px; }
30   </style>
31 </head>
32 <body>
33   <div feedbackForm>
34
35     {% if error %}
36       <div class="err">{{ error }}</div>
37     {% endif %}
38     {%- if success %}
39       <div class="ok">{{ success }}</div>
40     {% endif %}
41
42     <form method="post" action="{{ url_for('feedback') }}" novalidate>
43       <input type="hidden" name="csrf_token" value="{{ csrf_token }}">
44       <label>Name</label>
45       <input name="name" minlength="8" required value="{{ form.name }}">
46     </form>
47   </div>
48   <div feedbackList>
49     {%- for feedback in feedback_store %}
50       <div class="feedback">
51         <div>{{ feedback.name }}</div>
52         <div>{{ feedback.email }}</div>
53         <div>{{ feedback.message }}</div>
54       </div>
55     </for>
56   </div>
57 </body>
58 </html>
59 """
60
61 @app.route("/")
62 def index():
63     return render_template_string(PAGE)
64
65 @app.route("/feedback", methods=["POST"])
66 def feedback():
67     name = request.form.get("name")
68     email = request.form.get("email")
69     message = request.form.get("message")
70
71     if not NAME_RE.match(name):
72         return render_template_string(PAGE, error="Name is invalid. Only letters, numbers, dots, hyphens, and underscores are allowed. Max 20 characters.", success=False)
73
74     if not EMAIL_RE.match(email):
75         return render_template_string(PAGE, error="Email is invalid. Please use a valid email address.", success=False)
76
77     if not message:
78         return render_template_string(PAGE, error="Message is required.", success=False)
79
80     feedback_store.append({
81         "name": name,
82         "email": email,
83         "message": message
84     })
85
86     return render_template_string(PAGE, error=False, success=True)
```

	AAC A 20.5.py 1 Secure Feedback Form X → ↻ http://127.0.0.1:5000/feedback Feedback Name Kruthan Kiran Email kruthankiran@gmail.com Message Hello There Its Kruthan (2303A51404) Submit Recent Feedback - No feedback yet. ← → ↻ http://127.0.0.1:5000/feedback Feedback Feedback submitted successfully. Name Email Message Submit Recent Feedback Kruthan Kiran (kruthankiran@gmail.com): Hello There its Kruthan (2303A51404)	
	Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.	